

Crop Disease Classification Rubric

DS 4002 – Instructor: Livvy Yurish

Due: TBD

Submission format: Upload a link to your GitHub repository on Canvas

Individual Assignment

Preparatory Assignments – Everything you have learned thus far as a 2nd year data science student.

Why am I doing this? This is your chance to put your knowledge into practice through a real-world scenario you might face in future coursework or your professional career. It also offers a valuable opportunity to strengthen your analytical and technical skills while gaining hands-on experience with machine learning models, specifically random forests.

What am I going to do? You will obtain the dataset for this project from the following GitHub repository: <https://github.com/lyurish00/DS4002-CS3>. You will review the provided random forest model that classifies crop type, and then you will build your own random forest model to classify the disease present (if any) in tomato images. The output for each image will be one of the following: early blight, septoria spot, bacterial spot, late blight, mosaic virus, yellow virus, mold, or healthy. You will document your methodology and results in a PDF and place them, along with the code you develop, in your own GitHub repository. You will also create a README containing the GitHub file structure, instructions for running your code, and your references.

Tips for success:

- Experiment with hyperparameters, such as the number of epochs, batch size, n estimators, max features, max depth, and criterion to increase the accuracy of your model. Consider using a grid search to automate some of this experimentation.
- If your accuracy is low even after trying to optimize your model, don't worry! You can reflect on reasons the accuracy of your model might be low, such as overfitting/underfitting or model architecture, in your PDF report.

How will I know I have succeeded? You have met the expectations on this crop disease classification project when you follow the criteria in the rubric below.

Spec Category	Spec Details
Formatting	GitHub Repository: Make sure it includes the following - The tomato images dataset provided to you in the GitHub link - Your code source file - PDF report with your methodology and key findings - A README.md with project structure, instructions for replication, and references
Source Code File	Goal: Write well commented code that classifies the disease present in tomatoes as one of the following: early blight, septoria spot, bacterial spot, late blight, mosaic virus, yellow virus, mold, or healthy - Write your code in Python using Google Colab. - Turn in as a Python notebook (.ipynb) - Build a random forest model using sklearn. - Include comments that explain how your code works. Make sure to describe any preprocessing steps and hyperparameters. -
PDF Report	Goal: Explain your methodology and present the results (PDF format) Methodology section - explain the following: - Preprocessing steps (data augmentation, feature scaling, etc.) - Train-test split - Any model optimizations you applied (tuning hyperparameters) Results section - Explain the key findings - Interpret your model's performance using accuracy and loss. If your model's performance is low (below 60%), reflect on possible causes for this and how it could be improved in the future. - Explain any issues you encountered and how you overcame them - Include and discuss visualizations from your Python source file
README.md	Goal: Orient people to your repository and give credit to your references - Use markdown headers to structure the content

	- Section 1: Project Structure - Provide an outline of the files included in your repository. You should briefly explain the contents of each file. - Section 2: Instructions for replicating your results - Give explicit step-by-step instructions for reproducing the results of the case study. Explain where to find the data, which code should be run, and how to run it. - Section 3: References - All references should be listed; use IEEE Documentation style
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Acknowledgements: Special thanks to Jess Taggart from UVA CTE for coaching on making this rubric. This structure is pulled from Streifer & Palmer (2020).